

Lecture 7

Sampling Distribution and Confidence Intervals

Part 1

SAMPLING DATA

MOTIVATING EXAMPLE

- ◆ You work at a manufacturing company producing climbing ropes.
- ◆ All the ropes produced must satisfy certain strength conditions to be certified.
- ◆ The only way to test the strength is actually forcing the rope until it breaks and get a measurement.
- ◆ How can you ensure that your ropes are safe without breaking them all?

We need to get a small sample out of all the possible ropes we produce, and use the sample to deduce the safety of all our products.

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Sampling

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Statistical Inference

SAMPLING IS THE BASIS OF STATISTICAL INFERENCE

- Sampling consists on extracting or measuring a random subset of data coming from a larger population.
- Using this subsample of data, we would like to estimates features or statistics about the whole data.
- It is a form of prediction, (but instead of predicting the relationship between two variables, we are predicting some intrinsic properties of one single variable, for instance its average value).
- All our conclusions need to be given with a certain level of confidence. The larger the sample, the higher our confidence in the inference.

DATABASES VS PHYSICAL DATA COLLECTION

If we already have a lot of data, the process of sampling may consist simply on extracting a subset of our database (example: recommendation system in amazon)

However, for some studies we need to collect the data physically. For instance:

- Asking people on the street to test a new coffee brand
- Physically testing a manufactured piece

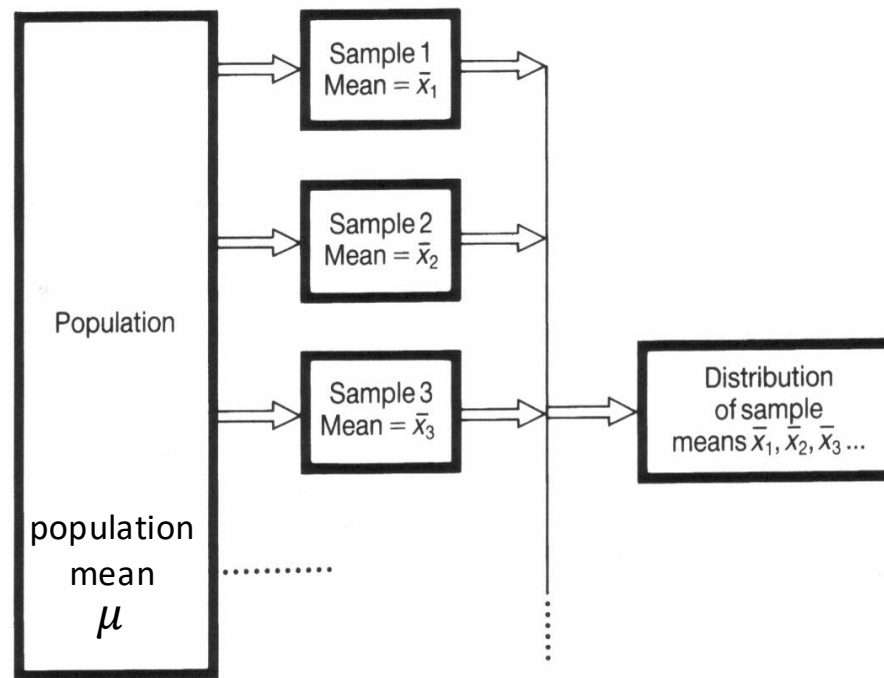
The way we collect this data is going to affect the quality and generality of my model

Make a subsample suitable for inference is much harder than it may look like. Let's look some sampling methodologies, and their benefits and potential biases

Method	Definition	Benefit	Potential bias	Example
Random Sampling	Random selection of sample	Easy to do. Cost efficient. Objective	Can ignore cases with low representation	make database query based on a random selection of id's.
Systematic sample	Data is selected at regular intervals	Simple. Representative	If there are cycles in the data, the sample can not be representative.	Select sales data for every Monday for the last 4 years. (maybe Monday not a typical day?)
Quota sampling	Data deterministic, but make sure the subsample has same quota for a number of characteristics	Make sure all subgroups are represented	Subjectivity when identifying what groups should be considered	When studying consumer behavior, identify profession of buyer, and make sure that the sample has the same proportions as the population.

SAMPLING FLUCTUATIONS

- ◆ For every sample that I extract, I will obtain different statistics and estimates.
- ◆ The variation is described by a **sampling distribution**



SAMPLING DISTRIBUTIONS. USEFUL FACTS

Consider a univariate population with mean μ and standard deviation σ . Let's take samples of size n . Then:

- If a population is normally distributed, the sampling distribution of the mean is also normally distributed
- Thanks to the Central Limit Theorem (**CLT**), with large samples (more than 30) the sampling distribution of the mean is approximately normally distributed regardless of the population distribution.
- In the two previous cases, the sampling distribution of the sample means has a mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$ (this quantity is also called **standard error, se**)

$$se = \frac{\sigma}{\sqrt{n}}$$

EXAMPLE IN PAPER

- ◆ Consider a dataset with all the weekly hours of Netflix consumption on European families. Suppose that this dataset has an average viewing time of 6 hours and a standard deviation of 1,2 hours.
- ◆ We collect many subsamples of 100 households each, and compute the average viewing time for each of the samples. What is the average of this subsamples? And the standard error?

Solution.

Mean of the samples must also be (close to) 6 hours

Standard error must be approximately $\frac{1,2}{\sqrt{100}} = 0,12$ hours